

SHAWBURY, United Kingdom

WMO: 034140

Lat: 52.7947N

Lon: 2.6647W

Elev: 236

StdP: 14.57

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.0	26.8	20.7	15.7	28.0	23.3	17.7	30.2	32.3	49.2	29.3	48.8	3.7	330	0.625

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	78.0	64.2	74.3	62.5	71.1	61.3	65.8	74.1	64.1	71.3	62.6	69.1	9.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.8	86.2	68.8	61.2	81.5	67.2	59.8	77.5	65.7	30.7	74.4	29.4	71.4	28.3	69.2	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.0	22.9	20.2	DB	16.8	83.4	5.2	3.6	13.1	86.1	10.1	88.2	7.2	90.2	3.4	92.9	(4)
(5)			WB	16.3	69.8	5.1	2.9	12.6	71.8	9.6	73.5	6.8	75.1	3.1	77.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	39.9	40.4	43.1	47.0	52.3	57.5	61.1	60.6	56.3	50.9	44.5	40.4	(6)
(7)		DBStd	9.36	6.08	5.83	5.34	5.28	5.19	4.54	4.47	4.23	4.21	5.41	5.94	7.06	(7)
(8)		HDD50	1496	314	271	220	118	33	2	0	0	3	54	180	301	(8)
(9)		HDD65	5670	778	688	678	540	393	229	139	150	262	438	614	761	(9)
(10)		CDD50	1338	1	3	7	27	106	227	344	328	193	82	15	5	(10)
(11)		CDD65	37	0	0	0	0	0	5	18	13	1	0	0	0	(11)
(12)		CDH74	345	0	0	0	0	9	57	176	92	9	2	0	0	(12)
(13)		CDH80	47	0	0	0	0	0	6	25	16	0	0	0	0	(13)
(14)	Wind	WSAvg	9.4	10.5	10.8	10.4	9.4	9.1	8.8	8.5	8.6	8.5	9.1	9.1	9.8	(14)
(15)	Precipitation	PrecAvg	25.9	2.2	1.7	1.8	1.9	2.2	2.1	2.0	2.4	2.4	2.4	2.4	2.5	(15)
(16)		PrecMax	34.1	4.3	4.4	3.8	4.5	6.8	4.9	4.7	5.0	6.9	4.5	5.3	5.1	(16)
(17)		PrecMin	17.9	0.4	0.2	0.3	0.3	0.5	0.5	0.3	0.3	0.1	0.4	0.9	0.4	(17)
(18)		PrecStd	3.6	1.0	0.9	0.8	0.9	1.2	1.1	1.1	1.1	1.5	1.1	1.1	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	56.5	60.6	69.2	75.2	80.5	83.1	82.3	75.1	66.8	59.7	57.3	(19)
(20)			MCWB	52.0	51.8	51.7	56.7	61.7	65.6	65.7	64.6	62.7	60.4	56.2	53.9	(20)
(21)		2%	DB	53.0	53.6	56.6	63.0	70.2	74.4	78.8	75.9	70.1	63.2	57.5	54.9	(21)
(22)			MCWB	49.9	49.9	50.4	53.4	59.3	62.2	64.3	63.8	61.3	58.3	54.4	52.0	(22)
(23)		5%	DB	51.0	51.6	54.1	59.3	65.8	70.3	75.0	71.9	67.0	61.2	55.7	52.9	(23)
(24)			MCWB	48.0	48.1	49.2	51.6	57.2	60.4	62.8	62.3	59.4	56.9	52.6	50.3	(24)
(25)		10%	DB	49.2	49.7	52.1	56.4	62.2	66.9	71.1	68.8	64.5	59.3	53.6	50.7	(25)
(26)			MCWB	46.5	46.3	47.7	50.1	54.9	58.7	61.4	60.8	58.0	55.5	50.5	48.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.5	52.4	53.7	57.5	63.6	67.2	68.7	67.7	65.3	61.5	57.1	54.0	(27)
(28)			MCD B	54.9	55.1	57.4	65.7	72.1	76.2	78.8	75.5	71.4	65.5	59.0	56.2	(28)
(29)		2%	WB	50.4	50.3	51.6	54.7	60.3	63.7	65.9	65.4	62.8	59.4	54.9	52.5	(29)
(30)			MCD B	52.7	53.2	55.4	61.5	68.3	71.2	75.2	72.5	68.2	62.5	57.1	54.7	(30)
(31)		5%	WB	48.5	48.5	49.9	52.7	58.1	61.5	64.1	63.7	60.8	57.7	52.9	50.5	(31)
(32)			MCD B	50.7	51.2	53.5	58.3	64.5	68.3	72.1	69.9	64.7	60.5	55.2	52.6	(32)
(33)		10%	WB	46.7	46.7	48.2	50.8	55.9	59.7	62.3	62.2	59.1	55.9	51.0	48.3	(33)
(34)			MCD B	48.9	49.4	51.6	55.6	61.0	65.8	69.6	67.7	63.4	58.7	53.2	50.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.0	11.2	13.3	15.5	16.0	15.9	16.3	15.6	15.0	12.8	11.0	10.1	(35)
(36)			MCDBR	11.3	12.1	16.0	20.0	21.9	22.6	23.5	20.1	18.9	13.4	10.8	10.6	(36)
(37)			MCWBR	10.4	9.7	11.1	12.0	13.0	12.4	11.5	10.1	10.9	9.4	9.3	9.8	(37)
(38)		5% WB	MCDBR	11.0	11.9	14.0	18.4	19.6	19.4	20.0	17.1	15.7	12.5	11.0	10.8	(38)
(39)			MCWBR	10.4	10.3	10.5	11.8	12.4	11.6	11.1	9.9	10.5	9.8	9.9	10.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.319	0.335	0.366	0.391	0.396	0.392	0.391	0.376	0.363	0.346	0.321	0.310	(40)	
(41)		taud	2.347	2.386	2.299	2.241	2.267	2.306	2.312	2.371	2.410	2.431	2.402	2.319	(41)	
(42)		Ebn at Noon	210	245	259	266	271	273	271	269	258	237	208	189	(42)	
(43)		Edh at Noon	19	25	33	39	40	39	38	34	29	23	18	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	221	429	788	1213	1518	1610	1508	1234	918	514	264	168	(44)	
(45)		RadStd	20	47	85	131	96	167	161	90	71	46	28	18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (77 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page